

Semester: VII					
INFROMATION STORAGE MANAGEMENT					
Category: PROFESSIONAL CORE ELECTIVE IV– (Group – F)					
(Theory)					
Course Code	:	CS374TFD		CIE	: 100 Marks
Credits: L:T:P	:	3:0:0		SEE	: 100 Marks
Total Hours	:	40L		SEE Duration	: 3 Hours
Unit-I					8 Hrs
Introduction to Information Storage and Management: Storage System Environment: Information Storage, Evolution of Storage Technology and Architecture, Data Center Infrastructure, Key Challenges in Managing Information, Information Lifecycle Components of Storage System Environment, Disk Drive Components, Disk Drive Performance, Fundamental Laws Governing Disk Performance, Logical Components of the Host, Application Requirements and Disk Performance. Storage provisioning (including virtual provisioning)					
Unit – II					8 Hrs
Data Protection, Intelligent Storage system: Implementation of RAID, RAID Array Components, RAID Levels, RAID Comparison, RAID Impact on Disk Performance, Hot Spares Components of an Intelligent Storage System, Intelligent Storage Array Object based storage and unified storage platform					
Unit –III					8 Hrs
Direct-Attached Storage, SCSI, and Storage Area Networks: Types of DAS, DAS Benefits and Limitations, Disk Drive Interfaces, Introduction to Parallel SCSI, Overview of Fibre Channel, The SAN and Its Evolution, Components of SAN, FC Connectivity, Fibre Channel Ports, Fibre Channel Architecture, Zoning, Fibre Channel Login Types, FC Topologies Three-site remote replication					
Unit –IV					8 Hrs
Backup Purpose, Backup Considerations, Backup Granularity, Recovery Considerations, Backup Methods, Backup Process, Backup and restore Operations, Backup Topologies, Backup in NAS Environments, Backup Technologies. Local Replication, Remote Replication: Source and Target, Uses of Local Replicas, Data Consistency, Local Replication Technologies, Restore and Restart Considerations, Creating Multiple Replicas, Management Interface, Modes of Remote Replication, Remote Replication Technologies, Network Infrastructure, Continuous data protection.					
Unit –V					8 Hrs
Securing the Storage Infrastructure, Managing the Storage Infrastructure: Storage Security Framework, Risk Triad, Storage Security Domains, Security Implementations in Storage Networking Monitoring the Storage Infrastructure, Storage Management Activities, Storage Infrastructure Management Challenges, Developing an Ideal Solution, Cloud Infrastructure for storage.					
Course Outcomes: After completing the course, the students will be able to:-					
CO1	Analyze various storage architectures along with data center infrastructure requirements.				
CO2	Identify physical and logical components of a storage infrastructure with RAID levels data protection mechanism.				
CO3	Evaluate Storage Area Network types with backup and replications technologies.				
CO4	Monitor and Manage Storage Infrastructure with Security Implementations.				

Reference Books	
1.	Information storage and management- Somasundaram, Gnanasundaram, Alok Shrivatsava, 2nd Edition, 2015, Wiley publishing ISBN 978-81-265-3750-1.
2.	Storage Networking-Real World Skills for the CompTIA Storage+ Certification and Beyond by Nigel Poulton, Publishers, SYBEX a Wiley brand, 2015: ISBN-13 : 978-8126557677
3.	Storage Networks Explained – by Ulf Troppens, Wolfgang Muller-Freidt, Rainer Wolafka, IBM Storage Software Development, Germany. Publishers: Wiley
4.	Storage Networks: The Complete Reference – Robert Spalding, 2003, Tata McGraw Hill, ISBN: 978-007224764
5.	Introduction to Storage Area Networks - Jon Tate, Pall Beck, Hector Hugo, Ibarra Shanmuganathan Kumaravel, Libor Miklas, 9th Edition , December 2017, IBM Redbooks, ISBN-13: 9780738442884.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. Each quiz is evaluated for 10 marks adding up to 20 MARKS	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks , adding up to 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (10) Designing & Modeling (10) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY+LAB)		100

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type of questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100